

Stakeholder

Product: budget diversity pairs, not demo depth

One-liner

For a single household skill you do NOT need 10k demos of one kitchen — you need ~ 32 varied scenes/objects $\times \sim 50$ demos ($\sim 1.6\text{k/skill}$) with a modern chunked visuomotor policy to reach $\sim 90\%$ zero-shot on new kitchens/objects.

Opposite of “just collect more demos.”

Who cares — who pays

Reallocate collector time to scene churn

Data ops / fleet leads

Reallocate collector hours from depth → scene/object churn (DROID-aligned).

BEHAVIOR / challenge teams

Per-skill floor under ~20k total demo budget (TENTATIVE global from OXE pack).

Foundation-model labs

OXE width helps, but per-skill coverage still needed for ZS without FT.

Hardware startups

UMI afternoon collection → deployable Franka skill is a sales demo.

Non-consumers

Need dexterous multi-minute skills; need multi-robot OXE; need lang-VLA task laws.

Product recipe — 32 × 50

~~~1.6k demos/skill · 1 unique object per environment · maximize envs~~

**32**

env-object pairs

**× 50**

demos per pair

**≈ 1.6k**

total demos / skill

**~90%**

zero-shot SR target

**1 obj/env**

at  $M \geq 16$  multi-obj negligible

**K≈50**

plateau · not 120+

Validated on Pour / Mouse / Fold / Unplug — Tab.1 SR 85–92.5%. Two new tasks: 4 collectors × one afternoon.

# Decision: churn scenes, don't deepen

App. G.2 — diversity wins even at matched total demos

## WRONG

10k demos  
same kitchen  
same mug

Depth without diversity →  
plateau / weak OOD

## RIGHT

32 kitchens  
× 50 demos  
1 object each

~1.6k total → ~90% novel  
kitchen+object

## CHECK

Per-env QA  
stratify SR  
don't trust mean

Tab.12: one env can be 40% while  
mean 85%

# Table 1 — what you buy with 32×50

Four tasks · ~90% average · afternoon scale for new skills

85.0%

Pour Water

92.5%

Mouse Arrangement

87.5%

Fold Towels

90.0%

Unplug Charger

Score ≠ SR alone: staged 0–3 pts/step. Blind multi-policy eval under identical inits. Real eval is expensive (>15k rollouts here).

# Fig. 5 — why diversity is the budget unit

Power-law gap shrinks with #envs/#objects/#pairs — not with demo dumps

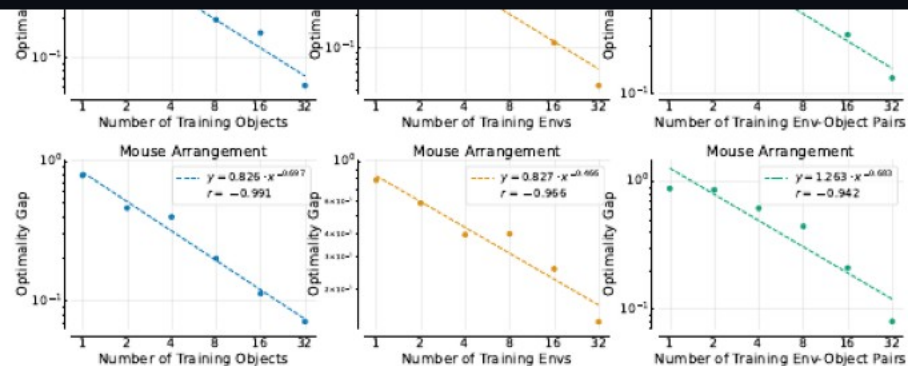


Figure 5: **Power-law relationship.** Dashed lines represent power-law fits, with the equations provided in the legend. All axes are shown on a logarithmic scale. The correlation coefficient (Pearson's  $r$ ) indicates a power-law relationship between the generalization ability and the number of objects, environments, and environment-object pairs. See Appendix G.1 for data scaling laws on MSE.

ing further emphasizes that expanding the diversity of environments and objects is more effective than merely increasing the number of demonstrations for each individual environment or object.

## 4.2 POWER-LAW FITTING AND QUANTITATIVE ANALYSIS

We next explore whether our experimental results follow power-law scaling laws, as seen in other domains. Specifically, if two variables  $Y$  and  $X$  satisfy the relation  $Y = \beta \cdot X^{\alpha}$ , they exhibit a power-law relationship. Applying a logarithmic transformation to both  $Y$  and  $X$  reveals a linear relationship:  $\log(Y) = \alpha \log(X) + \log(\beta)$ . In our context,  $Y$  represents the optimality gap, defined as the deviation from the maximum score (i.e.,  $1 - \text{Normalized Score}$ ), while  $X$  can denote the number of environments, objects, or demonstrations. Using data from our previous experiments with a 100% fraction of demonstrations, we fit a linear model to the log-transformed data, as shown in Fig. 5. Based on all the results, we summarize the following data scaling laws:

<sup>1</sup>Although we recognize the irreducible errors  $Y_{\infty}$  associated with scaling the data alone, fitting a three-parameter model  $Y = \beta X^{\alpha} + Y_{\infty}$  does not seem statistically justified given that we have only 6 data points.

# Risks & costs

Average ~90% hides 40% environments — env-stratify QA

## Env variance

Pour env#2 = 40% SR vs mean 85% — ship with per-env gates.

## Real eval cost

>15k rollouts; blind human scoring — do not replace with MSE.

## UMI SLAM limits

Texture-poor / large-occluder scenes dropped — skill family bias.

## Dexterity ceiling

Authors: harder dexterous/LH may need  $K > 50$  — recipe is for similar difficulty.

## Single policy stack

Laws measured inside DP+UMI — transfer to ACT/VLA unverified.

## Compute bar

Paper largest ~75 h on 8×A800 — plan scout mini-N first.

# Motion as a product feature

Buyer ask: chunked visuomotor + TE — not “just more data”

## DP chunks

visuomotor diffusion

## ACT TE

temporal ensemble stitch

## 5 Hz

Franka control

## Hor 16

action horizon

## Obs 2-3

history for pour/unplug

## Wrist RGB

UMI → Franka twin

# Contrast poles — pick the axis you can afford

This paper = per-skill env×object · not OXE robots · not DROID co-train alone

## This paper

Envs × objects × demos  
single-task DP  
ZS same-category

Power-law OOD; 32×50 recipe

## OXE

Embodiments × labs  
× skills

Foundation mix; FT transfer

## DROID

Scenes on one Franka  
co-train recipe

Scene diversity +22/+17

## UMI

Handheld interface  
in-wild locations

Collection mechanism used  
here

# Spine reminder

Five lines every stakeholder deck must keep

1. Diversity  $\gg$  demo depth — power laws on  $\#envs/\#objects/\#pairs$ ;  $K \approx 50$  plateau.

2. Recipe  $32 \times 50 \rightarrow \sim 90\%$  ZS;  $\sim 1.6k$  demos/skill — not 10k of one table.

3. Opposite “just more demos” — reallocate collector time to scene churn.

4. Tab.1 85–92.5%; ViT scales / U-Net doesn't; frozen/LfS fail;  $>40k/>15k$ .

5. Motion: DP + ACT TE @5 Hz; horizon 16 — product feature, not vibes.

# Numbers tattoo — pin these

Every deck carries the same spine numbers

**diversity  $\gg$  depth**

envs/objs/pairs  $\gg$  K

**32  $\times$  50**

pairs  $\times$  demos  $\rightarrow$   $\sim$ 90%

**85-92.5%**

Tab.1 SR  $\cdot$  4 tasks

**$\alpha \approx -0.47...-0.84$**

Fig.5 exponents

**0.66 / 0.81 / 0.90**

ViT-S / B / L

**0.88 / 0.90 / 0.83**

U-Net S / base / L

**0.00 / 0.03**

frozen / LfS fail

**K  $\approx$  50**

plateau / pair

**>40k / >15k**

demos / rollouts

**1191 pairs**

MA score 0.99 extrap.

**DP + ACT TE**

@5 Hz  $\cdot$  horizon 16

**FT 0.90 / LoRA 0.72**

full FT required

# Exam bite — Stakeholder

One sentence for the quiz / midterm

## EXAM BITE

Budget diversity pairs, not demo depth;  $32 \times 50$  is the published efficient frontier for single-task  $\sim 90\%$  OOD ( $\sim 1.6k$  demos/skill). Reallocate collector time to scene churn; env-stratify QA because averages hide 40% envs.

# Stakeholder — close

Takeaways + exam bite

1. Buy 32 varied (scene, object) pairs  $\times$   $\sim 50$  demos — not 10k of one kitchen.
2. Reallocate collector time to scene churn; multi-obj/env only helps when  $M$  is small.
3. Risk:  $\sim 90\%$  mean hides 40% envs — stratify QA; don't trust MSE.
4. Tattoo: diversity  $\gg$  depth  $\cdot 32 \times 50 \rightarrow \sim 90\% \cdot \text{Tab.1 } 85\text{--}92.5\% \cdot K \approx 50$ .

## EXAM BITE

Budget diversity pairs, not demo depth;  $32 \times 50 \rightarrow \sim 90\%$  single-task OOD.